ASSIGNMENT-II

1) Cellular Telephone Systems

→ Wireless Communication Systems

→ Divide a large geographical area into smaller areas called cells.

→ Each cell is served by a base station.

→ All base stations are interconnected through a central control unit called Mobile Switching Centre (MSC).

→ Allows a large no. of users to communicate simultaneously using radio frequencies.

Basic Structure

→ Cells: ★ Smaller regions the service area is divided into.
★ Hexagonal in shape.
★ Contains base stations.

→ Base Stations: ★ Located in each cell
★ Connects to mobiles within a cell
★ Handles radio communication
★ Transmits and receives radio signals from user devices.

→ Mobile Switching Centre (MSC): ★ Acts as a central control unit.
★ Manages
→ Cell setup
→ Call routing
→ Handovers
→ Billing
★ Connects to Public Switched Telephone Network (PSTN) and other MSCs.

→ Mobile Devices: ★ User's mobile phone / handset
★ Communicates with nearest base station
★ Equipped with unique identification for authentication and tracking.

Features

→ Frequency Reuse: * Each cell is assigned a group of frequencies.
* Non-adjacent cells reuse the same frequencies.
* Reduces interference, increases system capacity.

→ Handoff: * Performed by system when a mobile user moves from one cell to another.
* Call transferred seamlessly from one base station to another without dropping the call

→ Roaming: * Enables users to move between different services areas or networks while maintaining services.
* MSC handles
→ user authentication
→ service handover

Cellular System Operation

→ User places a call using a mobile device.
→ Signal is sent to nearest base station.
→ Base station forwards the call to MSC
→ MSC connects call to destination.
→ During movement, if the user enters a new cell, the system initiates a handoff.

Advantages

→ High capacity
→ Mobility
→ Scalability
→ Security and Privacy
→ Spectrum usage is efficient.

Applications

→ Mobile Telephony
→ Messaging
→ Internet Access
→ Emergency Communication
→ IoT Connectivity
→ Smart Device Networking

2) Flat and Frequency Selective Fading Channels in Communications

→ In wireless communication systems, transmitted signal may undergo various impairments due to transmission medium.

→ One such impairment is fading: variations in signal amplitude and phase due to multiple signal paths.

→ Classified into 2 types:

* flat-fading
* frequency-selective fading

Flat Fading

→ Occurs when coherence bandwidth > bandwidth of transmitted signal.

→ Also known as frequency-non-selective fading.

→ All frequency components experience same amount of fading.

→ Coherence Bandwidth: Statistical measure of range of frequencies over which channel has a constant or highly correlated response.

Characteristics

→ Channel behaves like a signal-tap filter.

→ No frequency distortion occurs.

→ Only amplitude variation are observed.

→ phase, amplitude change over time, but uniformly across frequency

→ Occurs commonly in narrowband systems, hence also called narrowband fading.

→ No inter-symbol interference (ISI)

Applications

→ Voice calls in GSM

→ Narrowband IoT Systems

Frequency Selective Fading Channels

→ Occurs when signal bandwidth > coherence bandwidth.

→ Different frequency components of the signal experience different levels of fading.

Characteristics

→ Acts like multi-tap filter with delays.

→ Suffers from distortion, leading to errors.

→ Typical in wideband systems, hence called wideband fading.

→ Suffers from ISI, which needs to be managed using equilisation or other techniques.

Applications

→ 3G, 4G, and 5G systems.

→ Orthogonal Frequency Division Multiplexing (OFDM) based systems.

3) Cordless Telephone Systems

→ Wireless communication ~~systems~~ devices.

→ Provide short-range voice communication btwn a portable handset and a base unit connected to PSTN.

→ Designed for limited mobility, typically within a home or office environment.

Basic Working Principle

(i) Base Unit: * Connected to telephone line
* Handles all routing and communication with the handset.

(ii) Handset: * Battery-powered portable device used to make or receive calls wirelessly.

Base and handset communicate using radio frequencies, commonly in the

→ 1.9 GHz
→ 2.4 GHz
→ 5.4 GHz bands.

Features

→ Short-range (50 - 300 m)
→ Full-duplex communication
→ Rechargeable handset battery
→ Caller ID, call waiting, answering machine integration.
→ No need for wired connection b/w handset and base.

Generation of Cordless Phones

(i) 1st Gen: → Analog
→ Simple RF communication
→ Prone to interference and eavesdropping

(ii) 2nd Gen: → Digital
→ Improved voice quality
→ Security and range

(iii) Modern Cordless Phones: → Includes VoIP support
→ Bluetooth
→ Intercom features

Digital Enhanced Cordless Telecommunications (DECT)

→ Standard for digital cordless phones
→ Primarily used in Europe and other parts of the world
→ Operates in 1.9 GHz frequency band
→ Secure and encrypted communication.
→ Supports multiple handsets and simultaneous calls.
→ Lower interference compared to analog systems.

Applications

→ Residential Telephone Systems
→ Office and small business internal communication.
→ Wireless intercom systems.
→ Hotel Room communication systems.

Advantages

→ Mobility
→ Convenience
→ Easy installation
→ Cost-effective for short-range communication
→ Multiple handsets can connect to a single base unit.

Limitations

→ Limited range compared to cellular phones

→ Interference from other wireless devices.

→ Can't operate during power outages.

→ Not suitable for long-distance mobility.

4) Small and Large Signal Fading Channels

Fading can be broadly classified into 2 types based on the causes and impact duration:

→ Large-Scale Fading

→ Small-Scale Fading

Large-Scale Fading

→ Gradual change in avg received signal power due to large objects like

* buildings
* hills
* terrain

over long distances or as mobile user moves.

Causes

→ Path loss due to distance from transmitter.

→ Shadowing from large obstacles.

Characteristics

→ Occurs over distances of 100s of m to km.

→ Affects overall signal strength.

→ Modelled using log-normal distribution.

Mitigation Techniques

→ Power control
→ Use of repeaters
→ Smart antenna systems

Small Scale Encoding

→ Rapid fluctuations of received signal over short distances / time periods due to multipath propagation.

Causes

→ Reflection from buildings, vehicles, etc.
→ Doppler shift due to relative motion.
→ Interference btwn multiple signal paths

Types

(i) Flat-Fading: All frequency components affected equally
(ii) Frequency-selective Fading: Different frequency components fade differently.

Characteristics

→ Occurs over distances of a few metres or less.
→ Can cause deep fades or signal distortion
→ Modelled using Rayleigh or Rician distributions.

Mitigation Techniques

→ Diversity (time, frequency, spatial)
→ Equalisation
→ Error-correction coding